SITE SAFETY PLUS

Construction site health and safety F: Specialist activities

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Supporting
INFORMATION

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STREET WORKS AND ROAD WORKS

Overview

In recent years the Health and Safety Executive (HSE) has expressed concern at the number of workers and members of the public killed or seriously injured through accidents related to road works. The New Roads and Street Works Act 1991 (NRSWA) provides a legislative framework for street works (including work carried out by public utility companies).

The NRSWA is supported by the *Safety at street works and road works* (SSWRW) Code of Practice (CoP). This chapter is an introduction to the comprehensive Code of Practice, with the content being provided as an overview to the legal requirements and guidance on a range of topics and issues that need to be considered when setting up temporary traffic controls.

1.1 Introduction

Road works will almost always bring the workers carrying out construction work into close proximity with vehicles (both construction plant and passing traffic). The safety of the workers and vehicle drivers or operators will depend upon the works being properly planned, carried out by trained and qualified workers and the behaviour of drivers in passing vehicles.

Highways England data revealed that there were around 330 incidents of abuse reported from September 2019 to October 2020, an average of nearly one every day.

There are around 38.6 million registered vehicles in Britain, making our roads some of the busiest in the world. They are also some of the safest. In terms of overall road safety, current national figures show casualties are at their lowest level for 40 years. Although this is promising, overall injuries to road workers on the UK's roads have risen.

In today's traffic conditions, the live carriageway of any highway is a dangerous place to work.

Drivers ignoring temporary speed limits is commonplace, leaving road workers even more vulnerable and with little to protect them from approaching vehicles.

80% of road workers have been physically or verbally abused by motorists and 40% of workers are abused on either a daily or weekly basis.

To help improve safety for workers on the highways, Highways England brought around an initiative to look at the health and safety for major road schemes. *Raising the Bar* was to identify best practice, raise standards and improve supply chain engagement within major projects.

Raising the Bar covers the following areas via support documents and guidance available on the Government website.

- Defining the minimum requirement and standards to be implemented on all major projects.
- Setting out 'desirable' elements, for consideration and implementation if deemed appropriate.
- Giving further guidance and supporting information on areas to be considered and applied where practicable.
- Continually reviewing working practices to reduce the risk to road workers, including the introduction of local targets, investigating layout changes and trialling new technologies.
- Reducing speed and improving compliance throughout road works, including the use of average speed detection.

www **Visit the Government website for more information on this initiative, as well as safe working guidance on working on roads and highways.**

1.2 Important points

- Works include any placement, connection or maintenance of temporary or permanent services, pipes, cables, sewers or drains that are laid or located within the carriageway or footway. This also includes forming new or temporary entrances and roads where they join an existing highway or footpath.
- All work should be carried out in accordance with the *Safety at street works and road works – A Code of Practice*.
- The Code of Practice (CoP), often referred to as the *Red book*, that supports the legislation must be complied with. The CoP is for works on all highways and roads, except motorways and dual carriageways with a speed limit of 50 mph or more. The CoP specifies minimum safety requirements for:
  – signage, lighting and guarding
  – methods of traffic control
  – controlling the speed of passing traffic
  – works near to tramways and railways, equipment and vehicles.

- Site-specific risk assessments must be carried out.
- Road users, whether in a vehicle or on foot, approaching the works on the road or footway, from any direction, must be able to understand exactly what route they must follow, when to stop or wait, when to give way and at what speed they should be travelling.
- Operatives who carry out work on the highway must be competent to do so, particularly anyone involved with setting up the traffic management layout on site, including positioning signage, lighting, cones, fences and barriers, and traffic control measures.

1.3 Legislative requirements

Three pieces of legislation specific to streets, roads and highways have an impact upon the way that works are carried out. The main relevant requirements are outlined below.

www **Copies of all legislation can be found online.**

1.3.1 New Roads and Street Works Act 1991 (NRSWA)

- Applies to works carried out in a street by an undertaker exercising a statutory right to inspect, place and maintain pipes, cables, sewers or drains, which are laid in the carriageway or footway. The term *undertaker* also covers holders of street works licences.
- Provides a legislative framework for street works by undertakers (including public utilities).
- Created a number of criminal offences that are triable in Magistrates' Courts (Sheriffs' Courts in Scotland). Amongst the 20 possible offences are:
  - failing to carry out works safely (such as signing, lighting and guarding)
  - not having the works controlled by supervisors and/or operatives with the relevant prescribed qualifications
  - failing to complete the contracted works within a reasonable timescale.

When a Local Authority acts as an agent to an undertaker or contractor to carry out work for an undertaker, the execution of the works is governed by the Act. The efficient co-ordination of street works is one of the most important aspects of street works legislation, benefitting street authorities, undertakers and road users alike.

The main objectives of the NRSWA are shown below.

- Ensure safety.
- Protect the structure of the street and the apparatus in it.
- Minimise inconvenience to people using a street, including a specific reference to people with a disability.

The Act comprises of the following parts.

- Part 1. New Roads in England and Wales.
- Part 2. New Roads in Scotland.
- Part 3. Street Works in England and Wales.
- Part 4. Road Works in Scotland.
- Part 5. General.
- Schedules.

1.3.2 Highways Act 1980 (England and Wales), the Roads (Northern Ireland) Order 1993 and the Roads (Scotland) Act 1984

- Was introduced to tackle congestion and deal with the management and operation of the road network.
- Places a duty on local traffic authorities to ensure the efficient movement of traffic on their road network and the networks of surrounding authorities.
- Gives authorities additional tools to better manage parking policies, moving traffic enforcement and the co-ordination of street works.
- Empowers the Highway Authorities to impose conditions on those working on the highway. It makes provision for licences for skips and scaffolds, and places responsibility for safety with the Highway Authority. Contractors **must** obtain permission before working on the highway.
- Applies to all work on the highway. The Highway Authority (or Roads Authority in Scotland) must be consulted and grant permission for works to be carried out. This applies to any works for road construction or maintenance purposes.

1.3.3 Traffic Management Act 2004

- Places a duty on Local Authorities to make sure traffic moves freely on their roads and the roads of nearby authorities.
- Places a duty on street authorities (such as County Councils) to secure more efficient use of their networks, to avoid or reduce traffic congestion and disruption and to regulate the use of the network.
- Requires a contractor who intends to work on the road to provide notice in advance of all works that directly or indirectly affect the highway network, effectively booking the road space. Minimum notice periods for different categories of work are contained within the NRSWA.

- Empowers Local Authorities to introduce a permitting scheme, which is designed to control the carrying out of specified works, in specified streets, in a specified area. In doing this it replaces the notice system under the NRSWA, whereby utility companies informed Highway Authorities of their intentions to carry out works in their areas.

www **For further information on the Traffic Management Act visit the website.**

1.4 Non-statutory guidance

There are two main documents (outlined below) that support temporary traffic management control.

1.4.1 Traffic signs manual

Published by the Department for Transport, the *Traffic signs manual* contents are instructional for roads, parking, road management and traffic signs.

The manual gives guidance on the use of traffic signs and road markings prescribed by the Traffic Signs Regulations and General Directions 2016, and covers England, Wales, Scotland and Northern Ireland.

www **The chapters can be downloaded from the GOV.UK website.**

Chapters 1 to 7 outline and include details of permanent road signs and road markings.

The most relevant chapter for this section is chapter 8, which gives guidance on road works and temporary situations where signage is required.

Chapter 1. Introduction. Introduction and an outline of the historical, functional and design aspects of signs.

The chapter includes sections dealing with the positioning and mounting of signs, and their removal with supports along with the specification on sign design and illumination requirements.

Chapter 2. Primary routes. Contains primary route destinations in England by region.

Chapter 3. Regulatory signs. Gives guidelines on the correct use of signs prescribed by the Traffic Signs Regulations and General Directions. These include prohibited turns, waiting and loading restrictions, bus and cycle lanes, and so on.

There is also a comprehensive section dealing with the signing of speed limits.

Chapter 4. Warning signs. Warning signs are used to alert drivers to potential danger ahead. They indicate a need for special caution by road users and may require a reduction in speed or some other manoeuvre.

Chapter 5. Road markings. Road markings serve an important function in conveying to road users information and requirements that might not be possible using upright signs.

They have the advantage that they can often be seen when a verge-mounted sign is obscured, and, unlike such signs, they can provide a continuing message.

Chapter 7. The design of traffic signs. How sign faces are designed. Dealing with signs that are designed for a specific requirement or location (such as directional information signs and signs relating to the control of on-street waiting, loading and parking).

Chapter 8 (Part 1). Road works and temporary situations – Design. Guidance for the design of temporary traffic management arrangements, which should be implemented to facilitate maintenance work or in response to temporary situations.

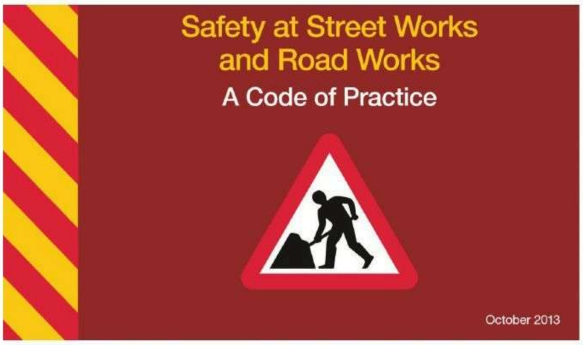
Chapter 8 (Part 2). Road works and temporary situations – Operations. Guidance for planning, managing and participating in operations to implement, maintain and remove temporary traffic management arrangements.

Chapter 8 (Part 3). Road works and temporary situations – Operations. Guidance for traffic safety measures and signs for road works and temporary situations.

Traffic control signage

1.4.2 *Safety at street works and road works – A Code of Practice* (CoP)

This updated CoP (commonly referred to as the *Red book*) came into force on 1 October 2014. The first printing (2013) contained a list of amendments for users to update manually. Readers are strongly advised to ensure they use the second printing, which incorporates all of the additional amendments: the version annotated '2nd impression (with amendments), June 2014'.



Check inside the cover to ensure you have the latest edition (the version annotated '2nd impression (with amendments), June 2014')

This CoP is a pocket-size reference book, intended to be used at the place of work. Its content is aimed at all workers on site, but it does not attempt to cover every situation.

Authorities with responsibility for road maintenance in England, Wales and Northern Ireland **must** comply with the CoP for their own works. In Scotland, the roads authority **should** comply with the CoP for their works, as recommended by Scottish ministers. Wherever the term *Highway Authorities* appears, it should be taken to refer to road authorities in Scotland and Northern Ireland, as well as in England and Wales. Failure to comply with the CoP is a criminal offence, potentially leading to prosecution and civil proceedings.

The CoP is applicable to all highways and roads, except motorways and dual carriageways with a speed limit of 50 mph or more. It does not apply to skips or scaffolding located in the highway.

Anyone planning to work on the highway must obtain and refer to the CoP as necessary. It is easy to use and should be incorporated into your own systems of work. It should form part of the safety management system for whoever is planning traffic management. Checking site set-up against the layout drawing is a quick and simple way to audit procedures. Whilst a supervisor qualified under the NRSWA will know what to do in most situations, it is the employer's responsibility to ensure that safe procedures are in place, followed and controlled under supervision.

Road users, including pedestrians, cyclists and horse riders, should not be put at risk and should be able to see the extent and nature of any risk before they reach it.

www **The Code of Practice can be freely downloaded from the GOV.UK website, or it can be purchased from TSO. For more detailed advice on subjects not covered in the Code of Practice refer to Chapter 8 of the *Traffic signs manual*, which can be freely downloaded.**

1.5 Carrying out road works

1.5.1 Before starting work

As with other construction projects, all road works fall within the Construction (Design and Management) Regulations 2015 and, as such, require a construction phase plan. Additionally, where more than one contractor is involved, a principal designer and principal contractor must be appointed, and the latter (or contractor, if only one) must actively manage site health and safety. Before starting work on the carriageway or footway, there are specific notice periods for informing the Highway/Streets Authority (or, in Scotland, the Roads Authority) that you intend to start, and three categories.

Notify before starting work

Minor. Three days' advance notice commencing with a duration of up to three days.

Standard. 10 days' advance notice commencing with a duration of four to 10 days.

Major. Three months' advance notice commencing with a duration of 10 days or more.

Depending on where the work is taking place, the Streets Authority may be the Local Authority, the County Council, the Department for Transport, or even the County Council on behalf of the Department for Transport, local Government and regions. The principal contractor or contractor should give notice to the recognised authority, taking into account any relevant lead-in period.

Lane rental schemes

The Department of Transport (DfT) has issued guidance to allow English local highways authorities to develop lane rental schemes. This gives local authorities the option to charge utility companies and highway authorities for digging up the very busiest streets at the busiest of times. The Government considers that charges must be applied only when works occupy the highway at peak periods, with exemptions from charges at other times, so as to provide a real financial incentive to carry out works at less disruptive times. After all initial costs have been covered, the remainder of the rental money is put back in road and future infrastructure investment.

1.5.2 Co-ordination of road works

When notice is given to the Highway Authority that the work is planned, they will inform the other utility undertakers in an attempt to co-ordinate works during a single excavation in order to avoid the same piece of roadway being continually excavated.

1.5.3 Major works

The term *major works* covers works carried out by the Highway Authority, which could include the following.

- The reconstruction or widening of the highway.
- Works on dual carriageways and at roundabouts.
- The construction of vehicle crossings over footways and verges.

1.5.4 Road widths

For two-way traffic, minimum road widths are shown below.

- 6.75 m on bus and HGV routes.
- 5.5 m on other roads for cars and light vehicles.

Any width less than 5.5 m is too narrow for two-way traffic and so the width must be reduced to a maximum of 3.7 m and a traffic control system introduced.

For shuttle working, minimum road widths are shown below.

Buses and HGV routes:

- 3.25 m desirable minimum width
- 3 m absolute minimum width.

Cars and light vehicles:

- 2.75 m desirable minimum width
- 2.5 m absolute minimum width.

1.5.5 Works on footways

An alternative safe route for pedestrians must meet the following requirements.

- Be provided before any footway or part of a footway is closed.
- Never be less than 1 m wide and ideally be 1.5 m or more in width.
- Be equipped with kerb ramps or a raised footway, as may be necessary, to take account of the needs of children, the elderly, people with disabilities (particularly visual impairment) or those with prams or wheelchairs.
- Be a non-slip surface and free of trip hazards (raised edges or overlaps on boarding may trip the elderly or infirm).
- Provide access to adjacent premises and public areas.

Kerb ramps to facilitate temporary pedestrian diversions

Footpath diversion showing the safe route for pedestrians

For further information refer to Appendix C and the CoP.

1.5.6 Excavations

In many cases the excavation of trenches will form a part of road works. It is essential that a safe working area is established before excavations begin and the following requirements are met for the duration of the works.

- Be adequately guarded, where necessary, to prevent the public or construction workers being put at risk of falls.
- Be supported where the risk assessment shows that it would be unsafe not to do so.
- Be inspected before every shift where people are entering the excavation to work.
- Be fitted with lights at night where otherwise construction workers or the public would be put at risk of falling.
- Be equipped with dewatering equipment where necessary.
- Be equipped with forced ventilation where a build-up of hazardous gases or vapours could occur.

Excavations must be adequately guarded

There may be occasions where trenchless technology (such as pipe-jacking or thrust-boring) would be more suitable for the work, although the use of such methods must first be cleared with the client and designers of the scheme.

For further details on safe working in or near to excavations refer to Chapter D06 Excavations.

1.6 Diversion and safety of apparatus

For further information refer to Chapter D07 Underground and overhead services.

1.6.1 Diversionary works

The term *diversionary works* covers works to:

- protect apparatus on site
- relocate apparatus elsewhere.

The *Safety at street works and road works* CoP contains comprehensive information on the responsibilities of the various utility undertakers and Highway Authorities relating to major road works that may affect the safety of apparatus and the need to divert apparatus.

The CoP includes guiding principles and outlines considerations that should be taken into account and the procedures to be followed.

Apparatus

Any pipe or ducting buried within the highway or pavement that is owned by one of the utilities. (Examples of apparatus are gas pipes, water mains, sewers, electricity cables, telephone cables and fibre optic communications cables.)

Unexpected damage to, or disconnection of, any apparatus could have serious implications for many people who are unassociated with the work taking place.

Construction work, which may put apparatus at risk, includes the following.

- The removal or construction of the footway or carriageway.
- Construction plant crossing or working in the vicinity of apparatus.
- The undermining or removal of side support to apparatus.
- Any deep excavation adjacent to apparatus.
- Piling or ground consolidation operations.

These risks can be minimised by providing suitable and safe vehicle crossing and access points, as outlined below.

- Temporarily moving or diverting apparatus to a safe location during the construction work.
- Protecting or temporarily supporting apparatus in situ.

Methods of supporting apparatus during excavation form part of the assessment process incorporated within the relevant operative and supervisor qualifications.

The correct construction of permanent or temporary vehicle crossings for site access or new road junctions is important to avoid putting apparatus at risk. The following issues need to be considered.

- The majority of service apparatus is located in footways.
- Footway construction layers must normally be excavated to accommodate thicker construction layers.
- The new construction may no longer provide adequate cover to apparatus.
- The vehicular loading may be greater than the apparatus can withstand.
- Vibrations from vehicles may weaken joints over a period of time.

1.6.2 Safety of gas apparatus

1.6.2.1 Depth of cover

The normal minimum depth of cover for gas mains operating in the low and medium pressure ranges is:

- 600 mm in footways or verges
- 750 mm in carriageways.

However, in practice these depths may vary, as each area gas distribution network operator can have its own standards.

In certain circumstances, depending upon the mains material, operating pressure and depth of cover, it may be acceptable for the mains to remain in situ when only subjected to light traffic (for example, a vehicle layby or crossing). It is not generally permissible to allow cast iron mains previously in the footway or verge to be subjected to vehicular traffic.

1.6.2.2 Risks during construction

Liaison with the gas undertaker during the planning stage is essential because existing mains, specifically older materials (such as cast iron), cannot be raised, lowered or moved laterally even by a few millimetres without risk. Gas apparatus must not be undermined and certain apparatus is particularly vulnerable when adjacent to deep excavations.

Any proposals to excavate near live gas apparatus should be discussed with the gas undertaker. Mechanical excavation must not take place within 0.5 m of low pressure gas supplies and distances should be agreed with the gas undertaker for medium and high pressure mains.

1.6.3 Safety of water apparatus

Water mains may be protected or diverted. Requirements will be determined by the work taking place.

1.6.3.1 Depth of cover

The standard minimum cover for water mains is 900 mm. However, there are three types of mains (trunk mains, distributor mains and service pipes) and the depth of cover may vary according to the type and installation.

Further information can be found in the CoP and by consulting the water company in whose area you are working.

1.6.3.2 Risks during construction

Some water mains are under higher pressures and are liable to rupture with a reduction in cover. If water mains are medium or high pressure, the Water Authority must be contacted and any work must be supervised by the Water Authority representative. The distance is usually between three to five metres either side of the pressured main.

Construction plant and heavy vehicles travelling over water apparatus with temporarily reduced cover can be an unacceptable risk. Therefore, diversions may be necessary unless protective measures are utilised.

Factors influencing the decision to divert water apparatus must include the following.

- The maintenance of the continuity of supply and the water quality.
- Material types and condition.
- The inability to raise, lower or slew pipes.
- The possible loss of ground support to pipes with the consequential risk of damage.

1.6.4 Safety of telecommunications apparatus

The need to exclude moisture from telecommunications cables and joints places constraints on the extent to which older cables can be moved during works.

For maintenance purposes there is the added need for vehicles to have access and be located at or near jointing chambers.

1.6.4.1 Preferred depth of cover

Varying depths of cover may be found with this type of equipment, depending on the types and design of the cables. As a rough guide, telecommunications cabling can be found at depths from 350 mm and television cabling at depths from 250 mm below a verge or footway and up to 900 mm in a carriageway.

1.6.4.2 Risks during construction

According to circumstances, the apparatus may be left in situ if ducts are adequately protected from construction plant and vehicles by the use of metal plates or tracks.

In some cases, it may be possible to accommodate small, temporary or permanent alterations in the line of a duct track by bodily slewing, raising or lowering a nest of ducts with the cables in situ.

1.6.4.3 Overhead telecommunications lines

Poles must be positioned to:

- minimise the risk of damage to overhead lines by vehicles
- be of minimum inconvenience to pedestrians
- avoid obstructing access to premises.

Road alterations may necessitate the replacement of poles if the clearance under the cables becomes inadequate.

Minimum heights above ground for overhead telecommunications lines are typically:

- 5.5 m at any point over a street
- 6.1 m on bus routes
- 6.5 m on designated roads.

1.6.5 Safety of electrical apparatus

The following factors should be considered when protecting cables in situ or diverting apparatus.

1.6.5.1 Underground cables

- The need to protect and support potentially hazardous services from mechanical impact, damage, strain and vibration during and after road works.
- A requirement to maintain the security of supply if alternative circuits are not available.
- The operating voltage of the apparatus.

The depth at which electricity cables or ducts are usually laid in the ground is determined by the need to avoid undue interference or damage.

Dependent on the type of cable and the power that it may be carrying, the depth of cover may vary from 450 mm up to 900 mm.

It is common to find electrical cables at much shallower depths of cover than these, particularly over bridges or culverts, and extreme caution must be exercised.

Plan – Locate – Dig (example of a shallow service)

In all cases, where the depth of cover is likely to increase or decrease, the apparatus owner must be consulted.

1.6.5.2 Overhead power lines

- The supports and stays of overhead power lines may have to be relocated.
- Earth wires from supports may have to be re-sited.
- Underground pilot wires may be associated with the route of overhead power lines (particularly where tramways are located).

The minimum height of overhead power lines above ground varies according to the voltage of the cable and as directed by the service supplier. The minimum height for lines carrying 33 kV is 5.2 m, 132 kV is 6.7 m, 275 kV is 7 m and 400 kV is 7.3 m.

Any temporary increase in ground level could affect the safe clearance between an overhead power line and the machinery.

Example of temporary goalposts to prevent contact with overhead power lines

www **Further information can be found in *Guidance Note GS6* on the HSE website.**

1.6.5.3 Risks during construction

- Additional protection or temporary diversion may be necessary to prevent damage to any apparatus during the construction stage.
- The hazards of accidental contact with electrical services by persons or machinery on site must be fully assessed. The installation of temporary goalposts may be required to prevent access by machinery with high reach attachments to areas where overhead power lines are located.
- Damage to overhead and underground cables can, in certain circumstances, cause widespread loss of electrical supplies for a long period.

1.7 Site layout definitions

Works area is the excavation, chamber, opening, reinstatement, or wherever work is taking place.

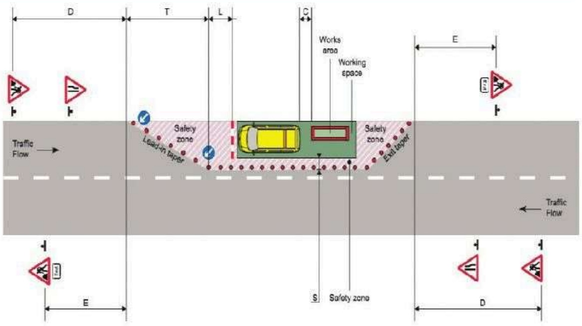
Working space is the space around the works area used to store tools, excavated material, equipment and plant.

It is also the space needed to move around the site without encroaching into live traffic or other hazards. Sufficient working space must be provided to ensure that the movement and operation of the plant (for example, swinging of jibs and excavator arms) is clear of passing traffic and is not encroaching into the safety zone.

Safety zone is the zone provided to protect site personnel from the traffic. No-one must enter the safety zone in the normal course of work.

It is only permissible to enter the zone to maintain cones and other road signs. Materials and equipment must not be placed in the safety zone.

The safety zone consists of the following.



Basic layout with a works vehicle

Notes

1. For numbers and minimum size of cones, and dimensions D, T, C, L, S and E, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

For a larger version of this image, please see Appendix B and Safety at street works and road works: a Code of Practice (2013).

Lead-in taper of cones (T). The length will vary with the speed limit and the width of the works.

Longways clearance (L). This is the distance between the lead-in taper of cones (T) and the working space. It will vary with the speed limit.

Sideways clearance (S). This is the width between the working space and moving traffic. It will vary with the speed limit.

For a chart showing the longways and sideways clearance dimensions required for works on roads with various speed limits refer to Appendix F.

Prosecution by the HSE following injury to a road worker

A utilities contractor was fined £12,000 after a worker was injured when they were hit by a bus passing through the contractor's roadworks.

The court was told that the safety zone was not being maintained and no other control measures were in place. The worker had to manoeuvre into the live traffic lane briefly to position some equipment as the bus was passing.

If an advisory speed limit is in operation the chart at Appendix F must be used to determine minimum longways and sideways clearances. Wherever traffic speeds are to be reduced, the method must be agreed in advance with the Highway Authority.

For safety reasons, it may be necessary to advise the emergency services of the location and duration of the works.

Working spaces and safety zones must be provided when personnel are present. If pedestrians are diverted into the carriageway, a safety zone must be provided between the outer pedestrian barrier and the traffic.

For further information refer to the CoP and also Appendix C.

Where the highway width is so restricted as to prohibit the provision of the appropriate sideways clearance and diversion of traffic would be impractical, traffic speeds must be reduced to less than 10 mph and a mandatory speed limit imposed.

There must also be an agreed safe method of working imposed on the site.

The safe system of work should be developed in advance and recorded in writing. It should prohibit working in the safety zone other than to maintain cones or road signs.

1.8 Signage and other site equipment

Traffic signs and other apparatus for the control of traffic must comply with the **Traffic Signs Regulations and General Directions**.

Compliance is achieved by the use of equipment conforming to the relevant British Standards. Equipment must also meet any requirements set out in the code as to size or performance.

All street works, irrespective of site and whether taking place at ground level or overhead, require adequate warning and information to be given to pedestrians and drivers.

For a selection of signs and equipment refer to Appendix A.

1.8.1 Temporary traffic management works

Consideration must always be given to operatives who are setting up, altering or removing temporary traffic management.

Clients and employers must ensure that operatives carrying out these operations are appropriately trained, supplied with the correct equipment and follow a safe system of work.

1.8.1.1 Crossing high-speed roads on foot

The HSE was so concerned about this aspect of temporary traffic management works that it published specific guidance on it, which applies to motorways and dual carriageways with a speed limit of 50 mph and above.

www **For further information refer to the HSE publication *Reducing risk in temporary traffic management operations* (CIS53).**

The guidance puts the initial responsibility on the designers of road works to ensure that the risks arising from the installation, alteration and removal of temporary traffic management controls are, where reasonably practicable, eliminated or reduced.

Any remaining significant risks should be highlighted and subsequently controlled by the careful planning and management of the works.

Systems of work should not rely on workers crossing the carriageway unless there are:

- adequate, unobstructed lines of sight
- traffic flows in which there are suitable gaps
- no more than four continuous lanes to cross (including slip roads)
- safe points to start and finish crossing.

When crossing the live carriageway, make sure that workers:

- are wearing appropriate personal protective equipment (PPE)
- can see and be seen by oncoming traffic
- can estimate appropriate safe gaps in the traffic
- stand back from the edge of the live carriageway before setting off
- set off only when a safe gap is present in the traffic
- walk straight across the carriageway
- carry equipment in a way that does not obstruct their view of oncoming traffic
- avoid running or zig-zagging between lanes
- avoid cat's eyes and other trip hazards
- carry signs and equipment in a way that minimises the risk of dropping them and reduces their resistance to wind
- avoid displaying the front of signs to oncoming traffic
- do not obscure their high-visibility clothing
- move to a position of safety after crossing (where possible, protected by safety fences or cones), at least 1.2 m from the live carriageway on high-speed roads and 0.5 m on all other roads
- take care when moving equipment so as not to endanger themselves or road users.

Managers and supervisors should judge whether to start work when poor weather conditions are forecast. If weather conditions deteriorate whilst work is in progress, the supervisor on site should decide whether or not to stop working.

Traffic management workers require high standards of physical fitness, eyesight and hearing. Operatives' health, fitness and suitability should be assessed before they are assigned to traffic management work.

Assessment should ensure that workers:

- have full, unrestricted use of their neck, trunk and legs
- have at least 6/12 distance vision when wearing glasses or contact lenses
- have good hearing
- are suitable for this work if they suffer from specific conditions (such as vertigo, balance disorders, gastrointestinal conditions and sleep disorders)
- are not taking inappropriate medication, illegal drugs or excessive amounts of alcohol.

Having 6/12 vision means you can see at 6 m what a person with perfect vision can see from a distance of 12 m.

1.8.2 Advance signs

A 'Road works ahead' sign must be placed where it can be easily seen, well before the works.

Its size and minimum distance from the start of the lead-in taper is governed by road type and traffic speed.

www **For further information refer to *Safety at street works and road works: a Code of Practice* (2013).**

A 'Road narrows ahead' sign, indicating which side of the carriageway is obstructed, should be placed between the 'Road works ahead' sign and commencement of the taper of cones.

1.8.3 Cones and lights

Traffic cones are placed to guide the traffic past the works.

The length of coned area and size of cones is governed by the speed limit and the type of road.

For further information refer to Appendix A.

Road warning lights are added at night, in poor daytime visibility and in bad weather. They must not be positioned higher than 1.2 m above the road surface.

www **For further information refer to *Safety at street works and road works: a Code of Practice* (2013).**

Cones and lights control road use during work

Road warning lights are used as shown in the table below.

Warning lights	Lights can be used		Lights must be used		
Speed limit	20 mph	30 mph	40 mph	50 mph	60 mph
Maximum height	1.5 m	1.5 m	1.5 m	1.2 m	1.2 m
Flashing option permitted (on street lit roads only)	Yes	Yes	Yes	No	No

1.8.4 Keep left and Keep right signs

Place 'Keep left' or 'Keep right' signs as appropriate at the beginning and end of the lead-in taper of cones.

1.8.5 Traffic barrier or Lane closed sign

A traffic barrier is positioned within the coned-off area facing oncoming traffic to indicate the width of the works site.

Lane closure signs are positioned ahead of the works area to warn drivers of the upcoming lane closure.

If a conspicuous works vehicle is present, a barrier may not be necessary *(for further information refer to 1.5.5 Works on footways).*

Managing closed lanes using good, clear signage

1.8.6 Pedestrian and traffic barriers

Pedestrian barriers should be used to mark out temporary footways, with a rigid barrier to protect pedestrians from traffic, excavation, plant or materials.

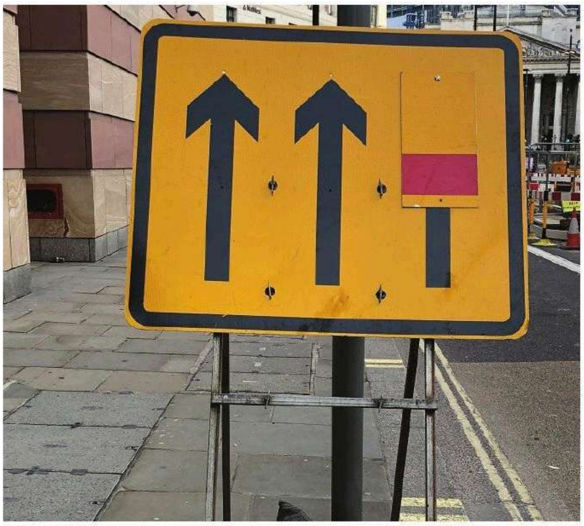
Handrails should be between 1 m and 1.2 m above ground level, with tapping rails (for blind or partially-sighted people) 150 mm deep, set with the lower edge up to 200 mm above ground level.

Road warning lights should be placed at the ends of the barriers at night. Barriers may be separate portable post and plank systems, gate frames linked together, or semi-permanent constructions built to enclose the site.

There are several different requirements for the barrier planks associated with post and plank systems.

The following section explains the requirements and how they may be met using barrier planks, which are red and white and manufactured in fully retro-reflective materials.

Barrier planks are required to carry out three functions, shown in the table below.



Sign showing right hand lane closure

1.	As a **traffic barrier**	When a traffic lane is closed, the regulations require this closure to be made using a retro-reflective* red and white barrier plank placed across the lane. This is illustrated as a traffic barrier in the top image, above. Advanced warning is shown by a 'Lane closed' sign (above).
2.	As a **pedestrian barrier**	Pedestrians must be separated from the works by barriers, which are conspicuous and mounted as part of a portable fencing system. Pedestrian barrier planks may be of several different contrasting colours; yellow, white or orange colours are best detected by partially-sighted people, but red and white is one of the acceptable combinations.
3.	As a **tapping rail** for blind and partially-sighted people	Tapping rails are placed as the bottom rail in a pedestrian fencing system. A red and white barrier plank may be used.

Note: *Since 1 April 2015 there has been an additional requirement that where a site is unattended for a period of 24 hours or more (including weekends), individual barriers or groups of barriers must be capable of withstanding winds of Class B (17.6 m/s) blowing from any direction.*

** Retro-reflective means that at night the material reflects light back to the light source.*

All barriers facing oncoming traffic should be of the fully retro-reflective red and white form. Red and white barrier planks do not have to be used for pedestrian barriers or tapping rails but, if they are, they must be retro-reflective. Other planks used for these purposes do not need to be retro-reflective. There are other points to note about the use of barrier planks in portable fencing systems.

- The traffic barrier ('Lane closed' sign) is not needed if the works are protected by a conspicuous vehicle.
- Pedestrian barrier systems must be rigid enough to guard pedestrians from traffic, excavations, plant or materials. They must be placed with sufficient clearance to prevent pedestrians falling into the excavation and, when placed to create a temporary footway in the carriageway parallel to the traffic stream, must be protected by a row of traffic cones between the barrier and the traffic stream. If the excavation is deep, or positioned close to pedestrians, stronger barriers may be needed and/or other safety measures may be required (such as covering or temporarily refilling the excavation).

  Where a work site may be approached by pedestrians crossing the road from the opposite side, you should place barriers, including tapping rails, all around the excavation, even when pedestrians are not diverted into the carriageway.
- Where long excavations are sited in situations where pedestrians are not expected to cross from the opposite side, barriers on the traffic stream side of the works area do not need the tapping rail. In these circumstances, on an unrestricted road, the barrier on the traffic stream side can be replaced with an additional row of cones. These cones should be linked with a suitably supported traffic tape to attract attention to the boundary of the safety zone.

All sides of an excavation where a pedestrian may gain access must be fenced off

Use pedestrian barriers to mark out any temporary footway. You must always use a rigid barrier to protect pedestrians from traffic, excavations, plant or materials. Place road warning lights at the ends of the barriers at night so that they may be clearly seen by pedestrians.

www **For further information refer to *Safety at street works and road works: a Code of Practice* (2013).**

1.8.7 Information board

An information board **must** be displayed at all sites, except mobile sites, short duration works and minor works that do not involve excavation. On sites where it is not a mandatory requirement, it is still desirable that a board is displayed.

As a minimum, the following information must be provided on the board.

- The name of the organisation undertaking the works.
- A telephone number for emergencies.
- The name of the principal contractor.

Wherever practical, the board should also contain other useful information, such as a description of the work that is being carried out, how long the work will take and a message apologising for any delay or inconvenience.

1.8.8 End sign

Road works that are 50 m or more in length (excluding the length of the tapers) must have a 'Road works ahead' sign and an 'End' plate, beyond the end of the works, to indicate the end of the work and any associated restrictions.

Where there is a series of two or more sites close together, the 'End' sign should only be placed after the last site (remember to cater to traffic travelling in both directions).

There is no need for an 'End' sign at the following locations.

- Where works are less than 50 m in length, unless there are two or more such sites close together.
- At road works on minor roads, with a speed limit of 30 mph or less, that do not carry a heavy volume of traffic or many large vehicles.

Where the permanent speed limit changes within the section of a road covered by the temporary speed restriction, any signs indicating the permanent speed limit must be obscured.

At the end of the temporary speed restriction, signs indicating the return to the permanent speed limit must be positioned on both sides of the carriageway, to indicate the speed limit from that point on.

www **For further information refer to *Safety at street works and road works: a Code of Practice* (2013).**

1.8.9 Sign lighting and reflectorisation

Signs and any plates used with them must be directly lit when all of the following conditions apply.

- There is a permanent speed limit of 50 mph or above.
- There is general street lighting.
- The street lighting is on.
- The sign is within 50 m of a street light.

All signs, including cones and red and white barrier planks, must be reflectorised to BS EN 12899-1:2007 Class 1 or Class 2. The only exception is the information board and pedestrian signs.

www **For further information refer to *Safety at street works and road works: a Code of Practice* (2013).**

1.8.10 Security and stability of signs

Only use signs that are approved and in accordance with the relevant British Standard, and ensure that they are placed correctly. All signs and guarding equipment should be properly secured so that they cannot be blown over by the wind or dislodged by passing traffic.

Built-in weights or sacks containing sand or other fine granular materials should be used, not kerbstones or similar weights, as they could be dangerous if hit by moving traffic.

Signs or guarding equipment should not be pushed into the ground as there may be unidentified services positioned below the surface. Make sure that the signs are correctly positioned and are of adequate size to give early warning of the hazard, in accordance with the table in the CoP.

Check signs regularly for position, stability and cleanliness as they may have fallen over, been moved, or become damaged or dirty.

Signs should be properly secured and checked regularly

1.8.11 Visibility of signs

All signs must be reflectorised, regularly cleaned to remove dust and dirt thrown up from passing vehicles and adequately lit after dark in accordance with the CoP. Signs must be placed so as to give the required forward visibility to approaching vehicles.

For sizing and siting distances of signage and cones refer to Appendix F.

A fallen sign means that drivers will be unaware of the hazard they are approaching and the actions they need to take

1.8.12 Two-way roads

On a two-way road, the signs should be set out for traffic in both directions.

1.8.13 Surplus signs

It is illegal to leave any signage or guarding equipment in position when it is no longer required. Signage and guarding that is no longer required should be removed immediately.

1.9 Traffic control

Select the method of traffic control from the table within the CoP. The CoP gives precise details of these methods.

www **You will also need to refer to the table in the inside back cover of the CoP (also included at Appendix F).**

For example diagrams on how to manage traffic control refer to Appendix B.

1.9.1 Traffic control by give and take system

Only use the give and take system when **all** of the following apply.

- The length of the works from the start of the lead-in taper to the end of the exit taper is 50 m or less.
- Two-way traffic is less than 20 vehicles counted over three minutes (400 vehicles per hour).
- Fewer than 20 heavy goods vehicles pass the site per hour.
- The speed limit is 30 mph or under.
- Drivers approaching from either direction can see 50 m beyond the end of the works.

Parking near the works, especially in front and opposite, is controlled or prohibited, unless visibility or lane width is unaffected.

1.9.2 Traffic control by priority signs

Only use priority signs when **all** of the following apply.

- The speed limit is 60 mph or less.
- Two-way traffic is less than 42 vehicles, counted over three minutes (840 vehicles per hour).
- The length of the works, including tapers, is no more than 80 m.
- Drivers approaching from either direction can see through the length of works from a specified distance before the works to a similar distance beyond the works. Distances vary according to speed restrictions, as follows:
  - 60 m on 30 mph roads
  - 70 m on 40 mph roads
  - 80 m on 50 mph roads
  - 100 m on 60 mph roads.

Priority over vehicles from opposite direction

1.9.3 Traffic control by stop/go boards

Remotely controlled stop/go boards should be used where possible. **All** of the following conditions must be met.

- The distance between stop/go boards is no more than 200 m.
- Use of the boards is restricted to daylight hours.
- An unobstructed view of both approaches is maintained.
- The operative is less than 100 m from both boards.
- Traffic flow is less than 840 vehicles per hour.

Where the shuttle lane with this type of system is more than 20 m, or it continues round a bend, stop/go boards will be needed at both ends of that lane. Manually-rotated stop/go boards can be used under the following circumstances.

Coned length (metres)	Maximum two-way traffic flow	
	Vehicles per three minutes	Vehicles per hour
Up to 100	70	1,400
101 to 200	63	1,250
201 to 300	53	1,050
301 to 400	47	950
401 to 500	42	850

1.9.4 Traffic control by portable traffic signals

Two-way portable traffic signals may only be used when **all** of the following requirements are met.

- The speed limit is a maximum of 60 mph.
- The length of the site is 300 m or less.
- The site does not straddle a railway level crossing.
- Traffic control is at least 50 m away from a level crossing equipped with twin red light signals.
- The Highway Authority has been informed and authorisation has been obtained.
- The signals are vehicle actuated (unless otherwise instructed by the Highway Authority).
- The signals are of a type approved for use on the highway.
- Stop/go boards are available in case the portable traffic signals cease to function correctly.

Traffic control in operation

Worker dies as a result of inadequate traffic control

A Magistrates' Court heard that workers from a sub-contractor were using a road planer to remove the old tar from the south bound side of the road, while the north bound side had traffic lights to control the direction of the traffic. During the activity one of the workers was hit by a passing vehicle. The man, aged 37, was taken to hospital, but died of his extensive injuries.

The HSE prosecuted both companies after an investigation found they had failed to ensure the road planing operation was carried out in a manner that allowed vehicles and pedestrians to move safely around the roadworks, without exposing persons to risks to their health and safety.

Speaking after the prosecution the HSE inspector said: 'In this instance the only control measures in place were cones along the centre of the road, and traffic was allowed to pass at 60 mph, close to the workers who were not provided with a safety zone because of the lack of space. Had adequate controls and a safe system of work been in place this terrible incident could have been prevented'. *(Source: HSE.)*

1.10 Personal protective equipment

1.10.1 High-visibility clothing

Given the nature of road works and the proximity of moving traffic, the importance of workers being clearly seen by colleagues and the travelling public cannot be overstated.

Whether working on site or just visiting, everyone must wear a high-visibility jacket or waistcoat at all times, manufactured to EN ISO 20471 standard requirements.

High-visibility clothing must conform to the relevant, current British or European standards and be worn as listed below.

- Correctly fastened.
- In a clean and usable condition.

Workers wearing PPE to Class 3 standard

- Where necessary, within the working space, as directed by the employer.
- When operating outside of the working space (such as when setting out, maintaining or removing signing, lighting, guarding and temporary traffic control).

The table below gives an indication of the requirements.

Class	Description	Colour
Class 1	Defines the lowest visibility level. The minimum level of protection required for any person working on a private road, or to be used in conjunction with high class garments. Minimum 0.10 m² of reflective material. Example: high-visibility trousers with two 5 cm reflective bands around each leg. These become Class 3 when worn with a Class 3 jacket.	The colour of the background should normally be fluorescent yellow or orange and the reflective material should comply with EN ISO 20471.
Class 2	Defines an intermediary visibility level. Required when working on or near to A or B class roads, and for delivery drivers. Minimum 0.13 m² of reflective material. Example: vest with two 5 cm bands of reflective tape around body or one 5 cm band around body and braces over both shoulders.	Colour scheme as Class 1.
Class 3	Defines the highest visibility level. Used when working near to or on motorways, dual carriageways or airports. Must be worn on dual carriageway roads with a speed limit of 50 mph or above. Minimum 0.20 m² of reflective material. Example: long-sleeved jacket and trouser suit with two 5 cm bands of reflective tape around the body and arms, and braces over both shoulders.	Colour scheme as Class 1.

High-visibility garments and jackets with sleeves must comply with Class 3 and be worn when working on or near live high-speed roads.

1.10.1.1 England and Northern Ireland

A risk assessment must determine the standard of high-visibility clothing required. In most cases, for work outside the working area, the likely minimum requirement is for a jacket with the maximum area of visible material, as specified in current British and European standards.

1.10.1.2 Scotland and Wales

In Scotland and Wales the general requirement is for a jacket to have full length sleeves and comply with current British and European standards. However, where a risk assessment indicates that full length sleeves would increase the risk of harm, due to the activity being carried out, three-quarter length sleeves may be worn.

1.10.2 Other personal protective equipment

The wearing of other PPE will invariably include safety footwear, which must have non-slip soles. Other PPE (such as head, hearing, eye and skin protection) must be worn, when required, depending on the nature of the work being carried out.

Hearing protection is not a requirement when crossing live carriageways. Hearing damage from passing traffic is considered to be less of a risk than a person being struck by a vehicle because they were not fully aware of what was going on around them. A risk assessment would be required in these circumstances.

For further information refer to Chapter B06 Personal protective equipment.

1.11 Training

Although the NRSWA refers to a trained operative, there is no specific requirement in the Act for operatives or supervisors to undergo training or produce evidence of prior training. However, it is a requirement of the New Roads and Street Works Act that at least one person on site at all times has a prescribed qualification as an operative for the task being carried out.

If the works involve breaking up or tunnelling under the street, an operative and a supervisor with prescribed qualifications for the task being carried out must be on site at all times. The Act also requires that the site is under the supervision of a qualified supervisor. The supervisor does not have to be on site at all times, allowing them to control more than one site at a time.

Furthermore, attention should be paid to the employer's duties regarding training as laid down in the Health and Safety at Work etc. Act 1974, the Management of Health and Safety at Work Regulations 1999 (as amended) and other relevant legislation. All Local Authorities will require evidence of the competence of operatives and supervisors prior to issuing a road opening or closure licence on the public highway.

Evidence of competence is achieved through the ownership of a street works card. Many contractors and individuals are registered on the street works qualification register (SWQR). To obtain the street works card individuals have to achieve one or more recognised qualification for the tasks they wish to carry out on site. These qualifications can be gained either through training and assessment or assessment only.

On achievement the body who award the qualification will contact the SWQR who, in turn, will register the individual and issue a street works card with the appropriate qualification endorsements. Each qualification is valid for five years. Two types of card are available, one at operative level and the other at supervisor level.

1.11.1 National Highways Sector Scheme 12

National Highways Sector Schemes (NHSS) are bespoke, quality management systems for organisations working on the UK road network. There are currently more than 20 individual schemes, covering a range of highways-related activities (such as fencing, landscaping, road surfacing and marking).

There are several parts to NHSS 12, which covers temporary traffic management. Each part relates to different road types *(shown right)*.

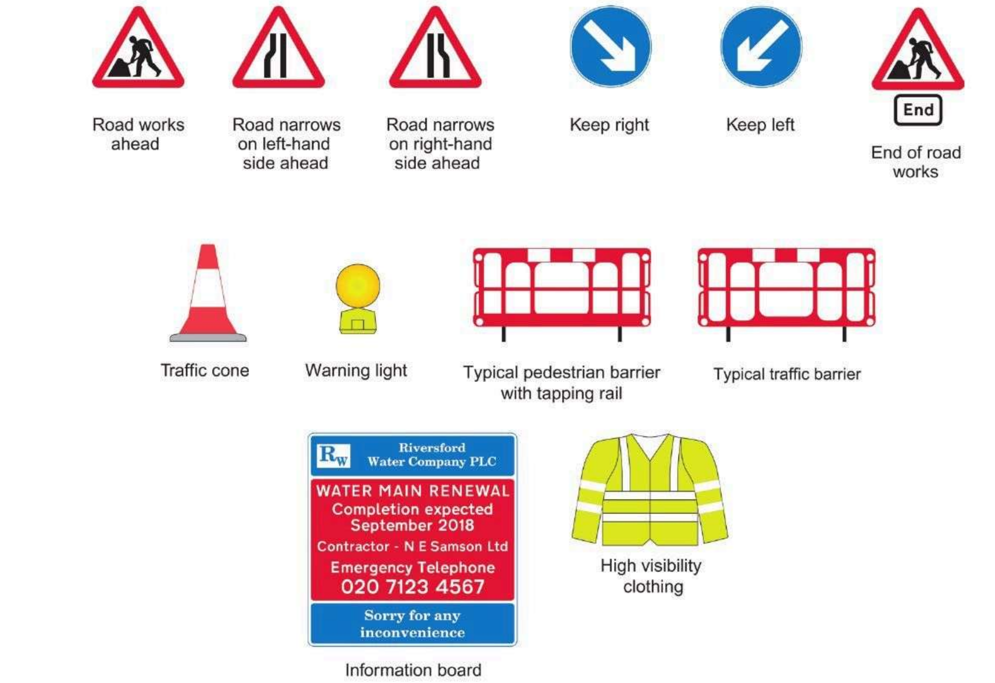
12A/B	High-speed motorways and dual carriageways.
12C	Mobile lane closures on motorways and other dual carriageways.
12D	Works on rural and urban roads.

Any company that supplies services to the Highways Agency under the Specification for Highways Works (SHW) must be certified under all the relevant schemes for the work they will carry out. The schemes aim to make sure that work is carried out to the highest standards of professionalism, using properly trained staff. The scheme places a strong emphasis on health and safety qualifications under the various schemes, which are awarded by Lantra Awards.

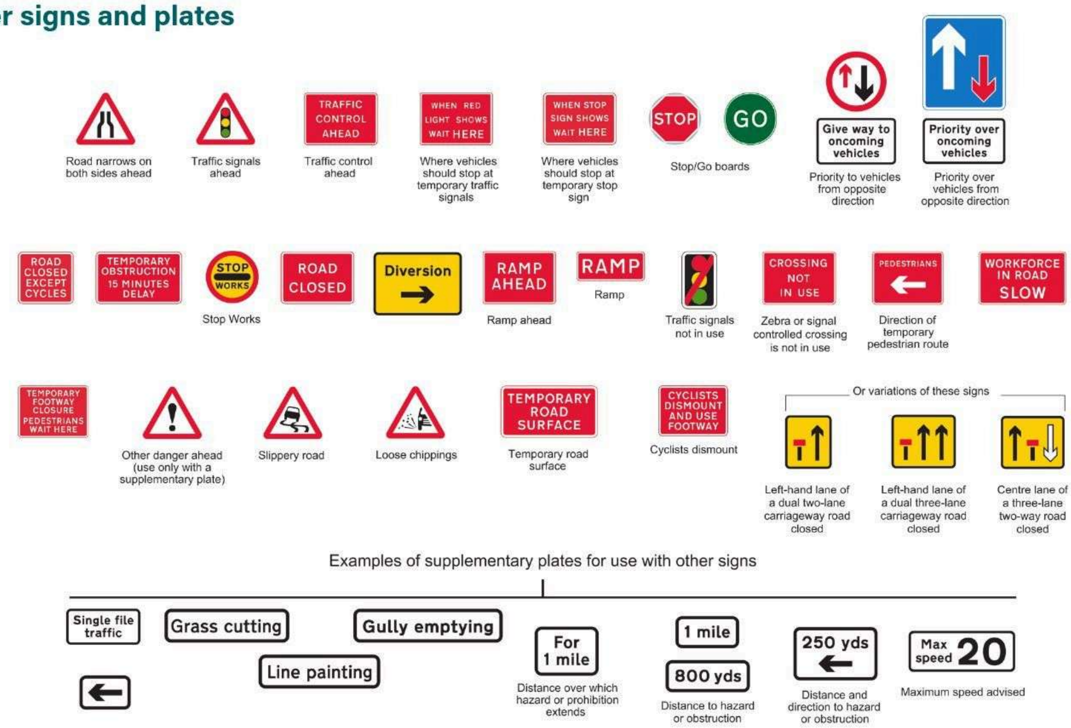
www **For the latest information, please visit *NHSS: certification for contractors and subcontractors* on the National Highways website.**

Appendix A – Signs and equipment

Basic signs and equipment



Other signs and plates



Note: *this shows some of the more common signs in use. It does not show every sign that might be required.*

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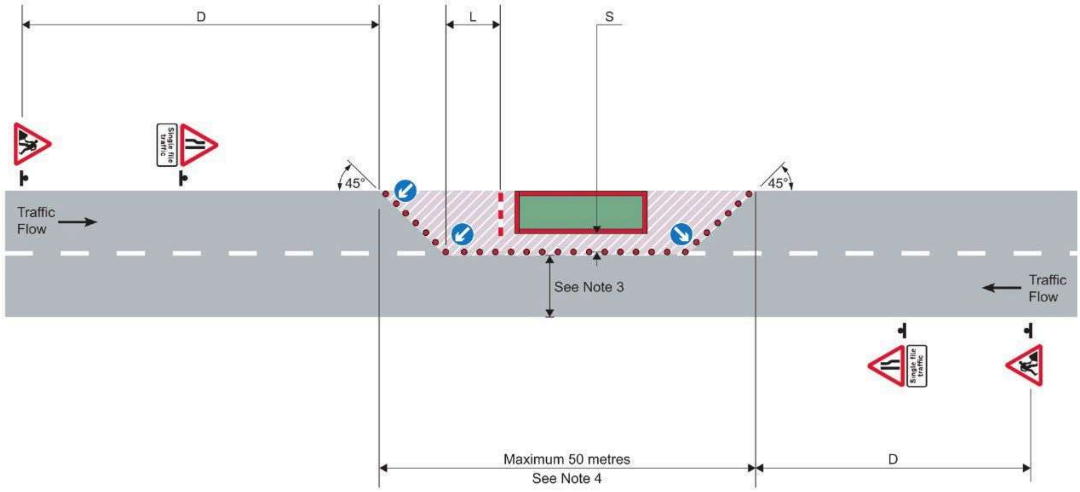
Appendix B – Traffic control

Traffic control by give and take for roads with a speed limit of 30 mph or less



Notes

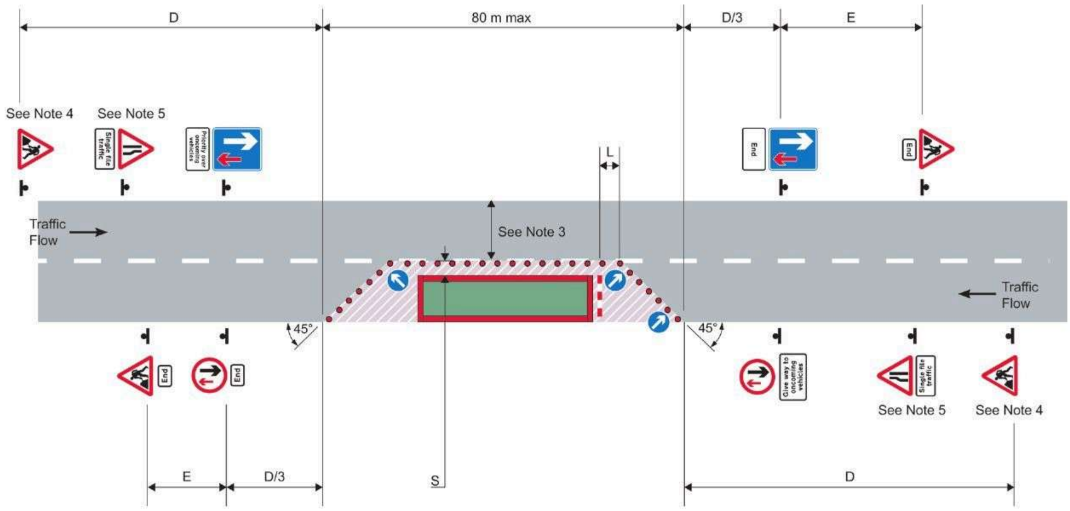
1. For numbers and minimum size of cones, and dimensions D, L and S, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

3. Refer to page 52 of the CoP for guidance on unobstructed width past the works.

4. 50 m maximum applies only where two-way flow cannot be maintained past the works.

Traffic control by priority signs



Notes

1. For numbers and minimum size of cones, and dimensions D, L, S and E, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

3. Refer to page 52 of the CoP for guidance on unobstructed width past the works.

4. A supplementary distance plate is required for roads with a speed limit of 50 mph or more.

5. For roads with a speed limit of 50 mph or more an assessment should be made as to whether 'for' and a distance should be included on a supplementary plate.

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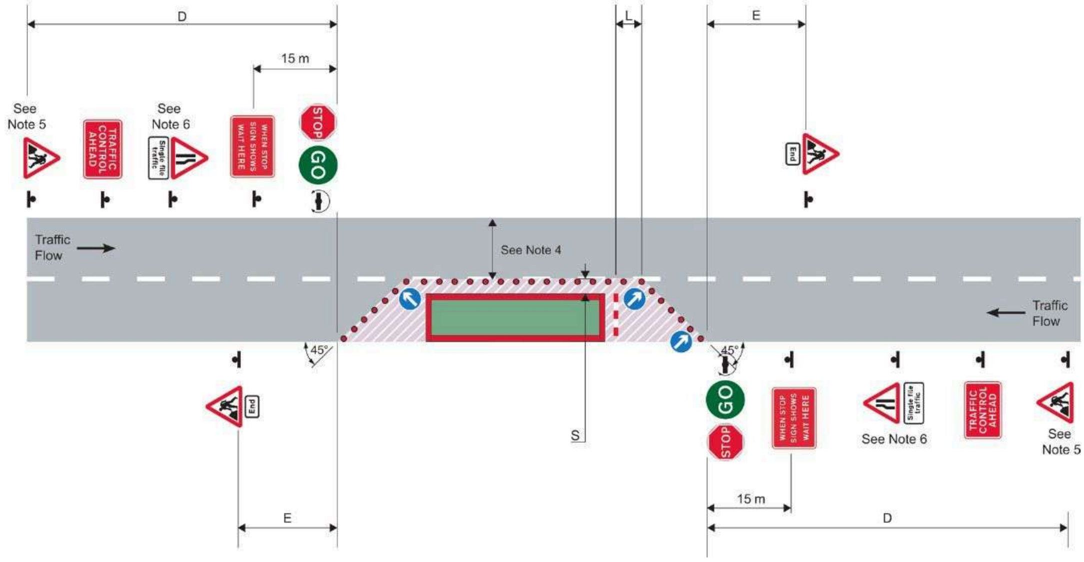
Traffic control by stop/go boards



Notes

1. For numbers and minimum size of cones, and dimensions D, L, S and E, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

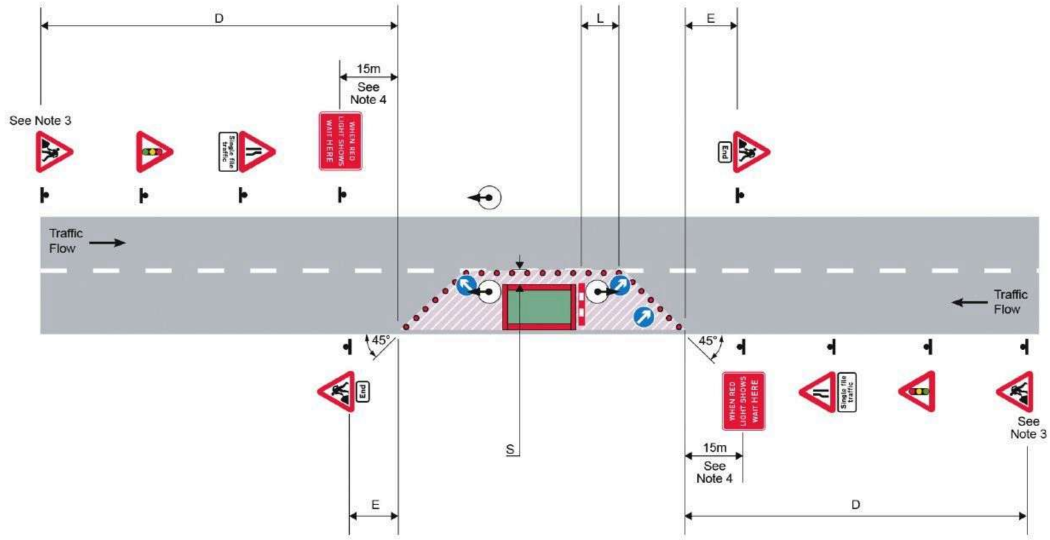
3. Stop/go boards should be placed where they will be in full view of approaching drivers. They may be located on either side of the carriageway.

4. Refer to page 52 of the CoP for guidance on unobstructed width past the works.

5. A supplementary distance plate is required for roads with a speed limit of 50 mph or more.

6. For roads with a speed limit of 50 mph or more an assessment should be made as to whether 'for' and a distance should be included on a supplementary plate.

Traffic control by portable traffic signals



Notes

1. For numbers and minimum size of cones, and dimensions D, L, S and E, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

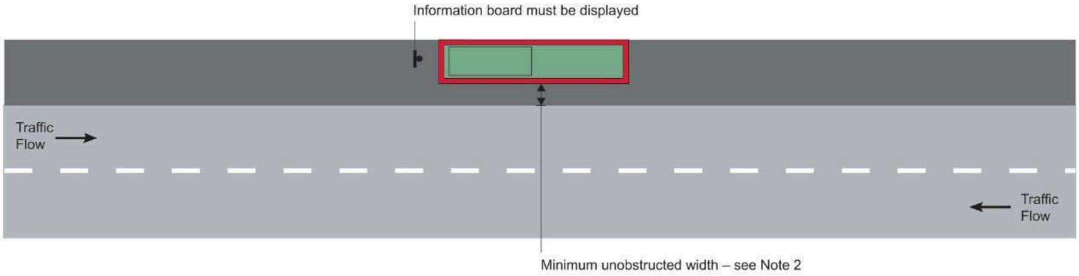
3. A supplementary distance plate is required for roads with a permanent speed limit of 50 mph or more.

4. This distance may have to be increased in some cases to allow larger vehicles to pass the works.

Based on © Crown copyright diagrams from *Safety at street works and road works: a Code of Practice* (2013). For further information visit the HSE website.

Appendix C – Works on footways

Works entirely on the footway

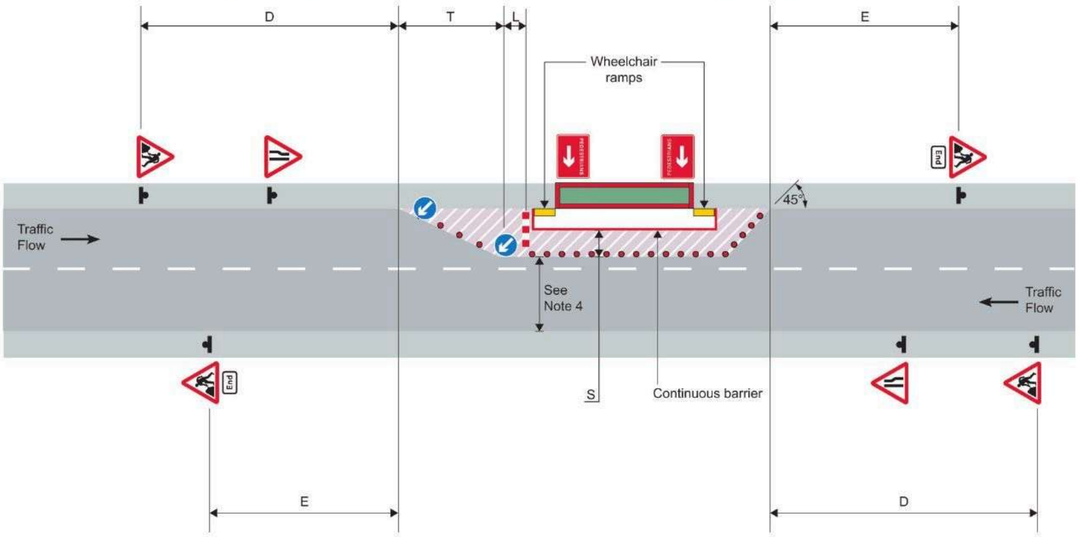


Notes

1. Advance signs are not required when the works, signing, lighting and guidance are entirely on the footway.

2. 1.5 m preferred minimum unobstructed width, 1 m absolute minimum.

Works on footway with pedestrian diversion into carriageway



Notes

1. For numbers and minimum size of cones, and dimensions D, T, L, S and E, refer to Appendix F.

2. An information board (omitted here for clarity) must be displayed.

3. Wheelchair ramps must be provided at the transition from footway to carriageway.

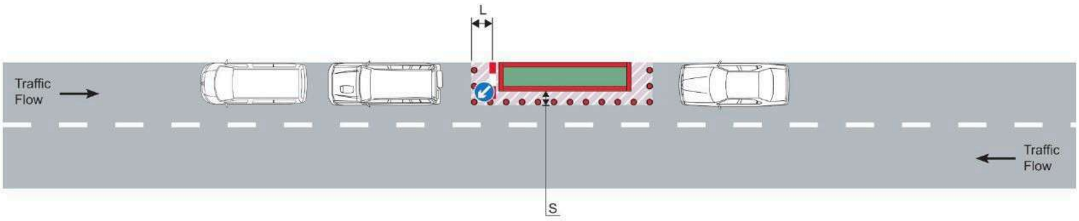
4. Refer to page 52 of the CoP for guidance on unobstructed width past the works.

5. Additional pedestrian barriers may be provided parallel or at right angles to the kerb, as site conditions require, to guide pedestrians past the works.

WWW **Based on © Crown copyright diagrams from *Safety at street works and road works: a Code of Practice* (2013). For further information visit the HSE website.**

Appendix D – Carriageway works

Works between parked vehicles with a speed limit of 30 mph or less



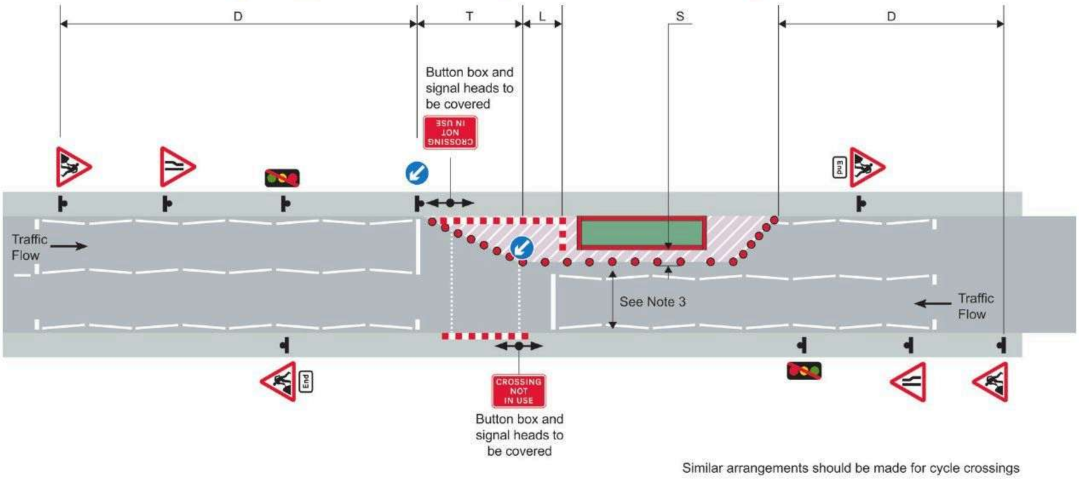
Notes

1. No advance signing, lead-in taper or exit taper required provided that the whole works, including the safety zone, do not extend beyond the line of vehicles.

2. An information board (omitted here for clarity) must be displayed.

3. If parked vehicles move away, tapers and advance signing should be provided.

Works obstructing a signal controlled pedestrian crossing



Notes

1. For numbers and minimum size of cones, and dimensions D, T, L and S, refer to Appendix F.

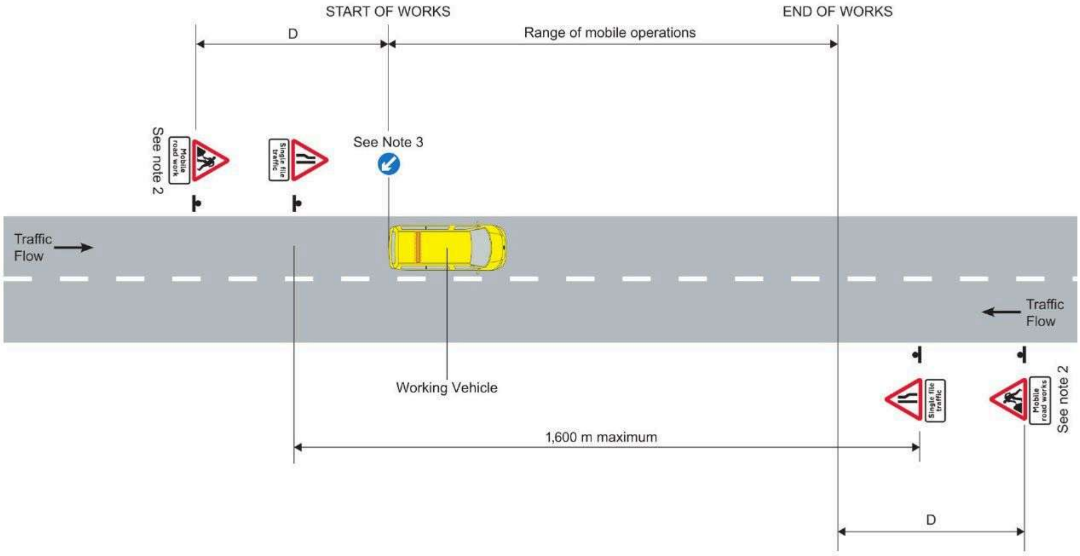
2. An information board (omitted here for clarity) must be displayed.

3. Refer to page 52 of the CoP for further guidance on unobstructed width past the works.

WWW **Based on © Crown copyright diagrams from *Safety at street works and road works: a Code of Practice* (2013). For further information visit the HSE website.**

Appendix E – Mobile works

Mobile works on a single carriageway road



Notes

1. For dimension D refer to Appendix F.

2. Or permitted variants.

3. Where a risk assessment determines that it is necessary to display signs on the working vehicle, they must comply with TSRGD (TSR (NI) in Northern Ireland).

www **Based on © Crown copyright diagrams from *Safety at street works and road works: a Code of Practice* (2013). For further information visit the HSE website.**

Appendix F – Setting out the site

(Distances in metres unless stated otherwise. Numbers are minimum numbers.)

Type of road	Minimum visibility distance to first sign	D Distance from first sign to start of lead-in taper	Lead-in taper	Width of works including sideways safety zone 1m	2m	3m	4m	5m	6m	7m	S Minimum width of sideways safety zone	E Distance from last cone to End of works sign	Minimum size of signs (mm)
Single carriageway – speed limit 30 mph or less	60	20 to 45	T Taper length	13	26	39	52	65	78	91	0.5	10 to 30	600
			No of cones	4	4	6	7	9	10	12			
			No of lights	–	–	–	–	–	–	–			
Single carriageway – speed limit 40 mph	60	45 to 110	T Taper length	20	40	60	80	100	120	140	0.5	30 to 45	750
			No of cones	4	6	8	10	13	15	17			
			No of lights	3	5	7	9	12	14	16			
Single carriageway – speed limit 50 mph or more	75	275 to 450	T Taper length	25	50	75	100	125	150	175	1.2	30 to 45	750
			No of cones	4	7	10	13	15	18	21			
			No of llights	3	6	9	12	14	17	20			
All-purpose dual carriageway – speed limit 40 mph or less	60	110 to 275	T Taper length	25	50	75	100	125	150	175	0.5	30 to 45	750
			No of cones	4	7	10	13	15	18	21			
			No of lights	3	6	9	12	14	17	20			

Speed limit mph	20	30	40	50	60
L Longways clearance	0.5	0.5	15	30	60

Speed limit mph	30 or less	40 or more
C Clearance to works vehicle	2	5

Notes

1. For roads covered by this Code, the minimum height of cones is 450 mm where the speed limit is 40 mph or less, and 750 mm where the speed limit is 50 mph or more.

2. The maximum spacing between cones in longitudinal lengths shall be 9 m, but no fewer than two cones shall be used in any length between tapers.

3. Lead-in tapers where two-way traffic control is used, and all exit tapers shall be at about 45° to the kerb line with cones spaced 1.2 m apart maximum.

4. In certain circumstances on congested roads with speed limits of 30 mph or less, the lead-in taper may be reduced to 45° (see page 19 of the CoP).

5. The longways clearance (L) is the distance between the end of the lead-in taper and the first traffic barrier placed across the lane.

WWW **Based on © Crown copyright diagrams from *Safety at street works and road works: a Code of Practice* (2013). For further information visit the HSE website.**

CONTENTS

Trackside safety

02

Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.

Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Working on a railway is a high-risk activity. Not only are there the normal problems of construction work to contend with, there is also the hazard of trains moving at high speed and close proximity. High Speed 1 was introduced for the Eurostar trains, which are capable of speeds of up to 186 mph. Domestic passenger services may run at 140 mph on this route. High Speed 2 (HS2) will enable line speeds of up to 250 mph. There could also be rail traction current supplied via overhead electricity lines at 25,000 volts AC or third and fourth rails carrying up to 750 volts DC.

Despite the UK having one of the safest railways in Europe, there has still been an increase of around 21% in workforce injuries. Some of these avoidable injuries were due to insufficient planning, failure to follow the safe system of work and worker competence.

It is important to understand that this chapter is intended as a brief guide for site managers to provide awareness only. For those who may be required to work, or put people to work, on the rail network, further information can be found in the Network Rail Standards and the *Rule book*.

www **Supporting information to complement this section can be found in the Rail Safety and Standards Board (RSSB) standards catalogue or the Network Rail Safety Central website.**

2.1 Introduction

Competence plays an important role in controlling health and safety risks when working on an operational railway system. It depends upon managing a complex mix of infrastructure, railway-specific rules, safety management systems and human factors.

Network Rail (which manages the railway infrastructure) has laid down standards and procedures that have to be strictly followed by all people working on railways. While the following information will contribute to working safely on railways, it can only be a summary of this complex subject and it is recommended that readers contact Network Rail directly for more detailed information relating to individual projects.

With the decentralised management of the railways, the fear was that the safety of the railway would be jeopardised by the introduction of independent railway operators. To prevent this happening the Office of Rail and Road (ORR) took overall responsibility for rail industry health and safety regulations in April 2006. At the same time new regulations were introduced in the form of the Railways and Other Guided Transport Systems (Safety) Regulations 2006 (ROGS Regulations).

The ROGS Regulations and the Health and Safety at Work etc. Act 1974 are administered and enforced by the ORR.

www **An explanation of how the ROGS Regulations apply to safety critical work is contained in the ORR publication *Safety critical tasks – Clarification of ROGS Regulations requirements* (Railways safety publication 4).**

The Rail Safety and Standards Board (RSSB) is a not-for-profit company, owned and funded by major stakeholders in the railway industry. A part of RSSB's remit is to publish and update Railway Group Standards, which govern the way that work is carried out on the railways, including the management of health and safety. Under the leadership of RSSB the industry has developed a series of *Rule book* handbooks, which are task based and cover a wide range of subjects, such as duties of the controller of site safety (COSS) or the safe work leader (SWL) and the practicalities, logistics and 'people' aspects of railway operations. The *Rule book* is also available as an app.

The primary purpose of the change to the handbook format was to rationalise the content and structure of the document so that it is more accurately targeted at the skill sets of end-users and clearly aligned with operational principles. The objectives of rationalisation and restructuring are explained below.

- Reduce rule-based errors, violations and misapplication.
- Enable end-users to exercise greater judgement and discretion in resolving operational issues.
- Reduce the need for, and the costs of, future rule changes.
- Support industry goals for competence management and performance improvement.

The onus is on organisations who work on the railways to make sure that they are referring to the most recent edition of any reference material.

Persons who work on the railways must achieve additional competencies beyond the requirements for those employed in conventional construction work.

For further details on the RSSB guide on developing the competence of staff at all levels to work on the rail network refer to Appendix C.

2.2 Important points

- There are two important terms you need to be aware of: 'Lineside' and 'On or near the line'. It is important that any work taking place on the lineside must not encroach or affect the area known as 'on or near the line'. Where any work is likely to affect the 'on or near the line' area, there must be a COSS or SWL present and operatives will require additional training.
- Means of access to the work site must be stated in the safe system of work (work package plan (WPP), task briefing sheet (TBS) or safe work package (SWP)). The access point may consist of a cabin, which may be staffed for the duration of the work.
- In order to ensure that members of the public do not gain access to rail sites, all sites must be secured at all times and any security or safety-related incident must be reported through the safety management information system (SMIS), as detailed in the Railway Group Standard GE/RT8047 – *Reporting of safety related information*.
- The site rules must be displayed in the access point; each member of staff is required to sign in and out of the site.
- The work area must be scanned using a Network Rail approved cable-avoidance tool (CAT) for underground services by a competent, authorised person prior to the work commencing. Any services identified must be marked and, if possible, temporarily removed (signalling cables).
- The COSS or SWL must brief the workforce on the site safety rules prior to the work starting, and brief the incoming COSS or SWL at the shift changeover.
- Following the briefing, each member of staff must sign the COSS or SWL briefing form to confirm they have understood the brief.
- Personal protective equipment (PPE) must comply with relevant legislation, British Standards and Railway Group Standards; the requirements are different to those for general non-railway construction work.

PPE requirements must comply with Network Rail's PPE policy.

Network Rail is constantly committed to eliminating all injuries and fatalities within the rail network industry. They run frequent safety campaigns to highlight specific areas of safety awareness. Visit their website to find out more about the latest campaign.

www **For further information visit the Network Rail Safety Central website.**

2.3 Network Rail licensing

www **If making enquiries about becoming a supplier of construction services to Network Rail, a list of the main contacts can be found on the Network Rail website, or you can contact them directly.**

All principal contractors who tender for work with Network Rail must maintain a principal contractor licence (PCL), which is subject to ongoing scrutiny by Network Rail.

When a contract is awarded, the principal contractor must produce a construction phase plan (CPP) for the contract. The principal contractor, their workforce and sub-contractors are then measured against the arrangements described in the plan.

A contractor will require sponsorship from Network Rail to enable them to enter the licensing system. The contractor applying for a licence will need to be able to demonstrate a significant level of knowledge of working in the railway environment.

The application process may, where appropriate, involve the requirement for specialist support.

However, it is important that everyone involved understands that the main aim is to check the contractor's existing systems, either against the Network Rail Standards or to align the contractor's systems with those of Network Rail (not to invent new ones).

Where a licence has been submitted, but is being reviewed, Network Rail may issue a provisional licence, which will allow the contractor to bid for rail work.

Acceptance of a licence by Network Rail should lead to training for the staff responsible for its implementation.

This is important because, whilst the document will contain the company's normal management systems, there will be a number of areas that are specific to rail work (such as drug and alcohol procedures, competence management, accident reporting and medical requirements).

Managers working on rail contracts must understand and implement these requirements.

2.3.1 Railway contractors certificate

The railway contractors certificate (RCC) will be required by companies who do not hold a principal contractor's licence but wish to employ and appoint a safe work leader (SWL) to act as the person in charge.

Holders of the RCC will be contracted directly to Network Rail, **not** the principal contractor.